Section 4.4: Final Exam Review

Finite Mathematics MTH 107 Fall 2014

Dr. James Ashe

Mars Hill University

Friday 14th November, 2014

/home/jim/JamesAshe/MHU/images/MarsHil

Schedule

http://www.mhu.edu/academics/course-catalog/exam-schedule

Final Exam	**Date**	**Time**
Section 07	Friday 5th December	8-10am
Section 10	Wednesday 3rd December	2:30-4:30pm

- Words into symbols

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Example 4.4.1.

We do our homeworkor we do not miss class.

- Words into symbols

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We do our homeworkor we do not miss class.

- Symbols into words

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We do our homeworkor we do not miss class.

- Symbols into words

Example 4.4.2.

Let h represent "We do our homework", and m represent we miss class". Symbolize: *If we miss class then we did not do our homework*.

- Negtaion

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  - All$\leftrightarrow$ some are not

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Example 4.4.3.

Negate: Some people are not going to have fun

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- Construct a truth table

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Example 4.4.3.

Negate: Some people are not going to have fun

- Construct a truth table

Example 4.4.4.

Construct a truth table for $(p \vee q) \rightarrow \sim q$

- Use a truth table to determine if an argument is valid.

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Example 4.4.5.

m: You like money.
f: You like food.

1. If you like food then you like money.
2. You do not like food.

Therefore, you do not like money.

		$\boxed{1}$	$\boxed{2}$		$\boxed{c}$	$\boxed{\text{Argument}}$
f	m			$1 \wedge 2$		
T	T					
T	F					
F	T					
F	F					

Chapter 2: Sets

- Basic set operations

- Basic set operations

Example 4.4.6.

Let $A = \{a, b, c, d\}$, $B = \{c, d, e, f\}$, $C = \{e, f, g\}$, and $U = \{a, b, c, d, e, f, g, \}$.

- Basic set operations

Example 4.4.6.

Let $A = \{a, b, c, d\}$, $B = \{c, d, e, f\}$, $C = \{e, f, g\}$, and $U = \{a, b, c, d, e, f, g, \}$.

- Find $A \cup B$

- Basic set operations

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- Find $A \cap B$

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- Find $A \cap B$
- Find $A \cap C$
- Find A'

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- Find $A \cap C$
- Find A'
- Find $(A \cup B)'$

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- Find $n(A)$

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- Find A'
- Find $(A \cup B)'$
- Find $A' \cap B'$
- Find $n(A)$
- Find $n(A \cup B)$

- Venn Diagrams

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Example 4.4.7.

A class has 50 students. 20 are biology majors, 30 are math majors, and 10 are math and biology majors. Construct and Venn diagram and label the number of students in each region

- Basic operations:

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Example 4.4.8.

Find the following:

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- When things can be repeated.

Fundamental Principle of Counting:

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A multiple choice test has 3 options for problems 1-3 and 4 options for problem 4-5. How many different ways can you select answers?

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- No repeats
- Order matters
- ${}_nP_k = \dfrac{n!}{(n-k)!}$

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How many ways can we choose 1^{st}, 2^{nd}, and 3^{rd} place in a beauty contest with 5 contestants?

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Example 4.4.12.

There are 10 balls. 6 blue and 4 red. Four balls are randomly selected. Find the probability that at least 2 are blue.

- $P(A) = \dfrac{n(A)}{n(U)}$

Find Simple Probability

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Example 4.4.13.

What is the probability that a $Q\spadesuit$ is drawn from a 52 card deck

Find Simple Probability

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- List of everything that can happen

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Example 4.4.14.

Two children are born; each is a girl or boy. Find the sample space

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Example 4.4.15.

What is the probability of having a boy and a girl?

With Venn Diagrams

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Example 4.4.16.

A candy bag has 25% caramels, 40% chocolates, and 15% chocolate caramels. If you randomly grab a candy, find the probability of the following:

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a. It is a caramel or a chocolate.

b. It is neither a caramel or chocolate candy.

c. It is just a chocolate candy.

Conditional Probability

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From a chart

Age	Obama	McCain	Other/NA
18-29	11.9%	5.8%	0.4%
30-44	15.1%	13.3%	0.6%
45-64	18.5%	18.1%	0.4%
65+	7.2%	8.5%	0.3%

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A car lot has 60% blue cars and the rest are red. Of the blue cars 25% are used and the rest are new. Of the red cars 80% are used and the rest are new.

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a. Draw a tree diagram.

b. Find the probability that a red car is new.

c. Find the probability that a randomly selected car is red AND it is new.

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Descriptive

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- Find histogram or frequency table.

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Chapter 4: Statistics

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When in doubt, DRAW it out.

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